

EE/RPF

For use from May/November 2027

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International Baccalaureate®
Baccalauréat International
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Student personal code:

Extended essay - Reflection and progress form

Student: This form is to be completed by you and your supervisor during the progress and completion of the extended essay. It records your progress through the extended essay process and includes your reflective statement addressing your extended essay experience. You must undertake three formal reflection sessions with your supervisor: The first formal reflection session should focus on your initial ideas and how you plan to undertake your research; the interim reflection session takes place once a significant amount of your research has been completed, and the final session will be in the form of a viva voce, once you have completed and handed in your extended essay. This document acts as a record in supporting the authenticity of your work. Following the viva voce, you write a **reflective statement** of no more than 500 words., which must be included on the RPF.

The completion of this form is a mandatory requirement of the extended essay. It must be submitted together with the completed essay. The reflective statement will then be assessed under Criterion E. Students and supervisors should be aware that a mark of 0 will be awarded by the examiner for criterion E if the RPF is blank or the reflective statement is written in a language other than that of the accompanying essay.

Supervisor: You must have three reflection sessions with each student, one early in the process, an interim meeting and then the final viva voce. Other check-in sessions are permitted but do not need to be recorded on this sheet. After each reflection session you must record the month that the reflection session took place, the student's year of DP study at that time and initial the form.

By submitting this student work for assessment, you are taking responsibility for its authenticity. No piece of student work should be uploaded/submitted to the e-Coursework system if its authenticity is in doubt.

First reflection session

Month: DP year (1 or 2): Supervisor initials:

Interim reflection session

Month: DP year (1 or 2): Supervisor initials:

Final reflection session – Viva voce

Month: DP year (1 or 2): Supervisor initials:

Reflective statement

(No more than 500 words, written in the same language as the accompanying extended essay)

I recently watched a video about how the enigmatic value of π was discovered and how it is still a mystery to us. I became curious about this matter and tried to find out more information about this irrational number. After completing a thorough examination, I learned that there are numerous ways to evaluate π . The earliest goes back to 200BC.

The most intriguing ancient method I came across was Archimedes' usage of regular polygons inscribed and circumscribed in a circle, this idea has been around for a very long time and has a significant influence on modern ways.

I next sought for software that would allow me to compute the value of π using the formulas I had read about. In contrast, I came to the conclusion that Microsoft Excel would be the best application to use because it is easily accessible and gives the accurate value.

As I embarked on my journey to find the value of π , I encountered several obstacles along the way. My initial attempts to use Java or Python scripts to calculate the approximation were met with disappointment as the code failed to yield accurate results. I was forced to turn to Excel, a tool that I had underestimated in its capabilities. Though it had its limitations, I was determined to make it work.

As I delved deeper into my research, I discovered two formulas – John Wallis series and Francois Viète's series – that offered a glimmer of hope. These formulas allowed me to use Excel more effectively, despite its limitations, and opened up a world of possibilities. My persistence in seeking alternative solutions was rewarded as I was able to complete my extended essay with confidence.

This experience was a testament to the power of determination and the importance of seeking out alternative solutions. It has taught me to never give up, even when faced with seemingly insurmountable challenges. The journey was not easy but the reward was sweet as I was able to unravel the mystery of π and find its true value.

I learned a number of skills while writing my extended essay, including how to conduct research and learn about very antique methods of calculating the value of π and how to do it using Microsoft Excel. I started this extended essay with very little knowledge of how the value of π is calculated, but this essay has taught me a lot of new ideas that I didn't know before and I gained mathematical knowledge. By making spreadsheets for each of these approaches I not only learned about them but also obtained a full grasp of how they are created. I was able to apply the mathematical concepts I had acquired in school and deepen my understanding of them by exercising them in a new way when proving the formulas in this article. This experience has left a lasting impression on me and has inspired me to continue learning and growing in new and exciting ways.

(498 words)